

Gradient

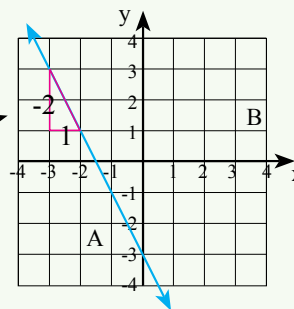
Equation of a line:

$$y = mx + c$$

m = gradient

c = intercept on y-axis

$$m = \frac{\text{rise}}{\text{run}} = \frac{-2}{1} = -2$$



$$y = -2x - 3$$

$$m = -2$$

y-intercept = -3 {cuts y-axis at (0, -3)}

Exercise 16.5

1 Find the gradient and the y-intercept of each of the following linear equations:

a) $y = x + 2$

b) $y = x - 3$

c) $y = 2x + 1$

d) $y = 2x - 1$

e) $y = -2x - 5$

f) $y = -x + 7.5$

g) $y = 1.2x - 2$

h) $y = -0.5x - 3.6$

2 Rearrange each of the following linear equations into the form $y = mx + c$ and then find the gradient and the y-intercept:

a) $y - x = 2$

$$2y + 3x = 9$$

b) $y + x = 3$

c) $y - 2x = -3$

$$2y = -3x + 9$$

d) $y + 3x = -4$

e) $2y + 3x = 1$

$$y = 1.5x + 4.5$$

f) $2y - 5x = -1$

g) $5y + 4x - 1 = 0$

$$m = 1.5$$

$$y\text{-intercept} = 4.5$$

h) $3y - 2x - 6 = 0$

3 For each of the following linear equations, find the gradient, the y-intercept, and the x-intercept. Use this information to sketch each linear equation.

$$y = -2x + 1$$

$$y = mx + c$$

$$m = -2, c = 1$$

$$\text{gradient} = -2$$

$$\text{y-intercept is at } (0, 1)$$

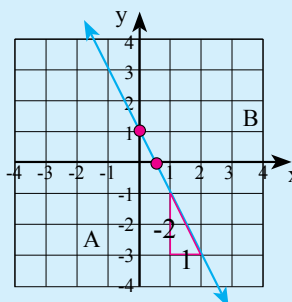
cuts the x-axis when $y=0$,

$$0 = -2x + 1$$

$$-1 = -2x$$

$$-1/-2 = x$$

$$0.5 = x \text{ thus } \text{x-intercept is at } (0.5, 0)$$



a) $y = x + 2$

b) $y = x - 3$

c) $y = 2x - 1$

d) $y = 2x + 4$

e) $y = -x - 2$

f) $y = -x + 5$

g) $y = 1.5x - 3$

h) $y = -1.2x - 3$

i) $y + 2x = 1$

j) $y - 3x = -3$

k) $3y + 1.5x - 4 = 0$

l) $2y - 4.2x + 3 = 0$

$$m = 1 \quad \theta = 45^\circ$$

$$m = 0 \quad \theta = 0^\circ$$

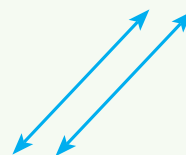
$$m = -1 \quad \theta = -45^\circ$$

Exercise 16.6

1 Show that the following pairs of lines are parallel:

- a) $y = 3x + 1$ and $y = 3x - 2$
- b) $y = 2x - 3$ and $3y - 6x + 1 = 0$

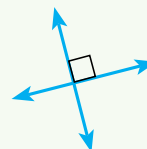
PARALLEL
lines that never cross
 $m_1 = m_2$



2 Which of the following pairs of lines are perpendicular:

- a) $y = -2x - 1$ and $y = 0.5x + 3$
- b) $y = x - 3$ and $2y - x + 3 = 0$
- c) $3y + 6x - 1 = 0$ and $2y - x + 3 = 0$

PERPENDICULAR
 90°
 $m_1 \times m_2 = -1$



3 What is the value of m if the line $y = mx + 1$ is:

- a) parallel to $3y + 2x - 4 = 0$?
- b) perpendicular to $3y + 2x - 4 = 0$?

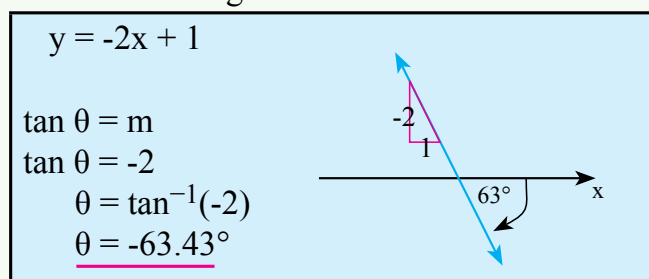
4 Are the following three points collinear:

- a) A(2,3), B(5,4), and C(8,5)
- b) A(5,4), B(-1,2), and C(-4,1)
- c) A(-1,-1), B(-3,-3), and C(6,6)

Points are collinear if they are on the same line.

The points A, B, C are
collinear if $m_{AB} = m_{BC}$

5 Find the angle at which each of the following lines cut the x-axis:



- a) $y = x + 3$
- b) $y = 2x - 5$
- c) $y = -2x + 1$
- d) $y = 5x - 2$
- e) $2y + 3x - 1 = 0$

6 If the gradient of the line segment A(3,-2), B(x,5) is 1, what is the value of x ?

7 If the gradient of the line segment A(1,4), B(x,-2) is 0.5, what is the value of x ?

8 What is the value of a if A(-1,3), B(2,0), and C(a, 4) are collinear?

9 What is the value of b if A(3,4), B(-3,0), and C(-2, b) are collinear?

10 What is the equation of the line that cuts the y-axis at (0, 2) and is parallel to the line segment A(-1,3), B(2,-3)?

11 What is the equation of the line that cuts the y-axis at (0, -3) and is parallel to the line segment A(2,-1), B(4,-2)?

12 What is the equation of the line that cuts the y-axis at (0, 1) and is perpendicular to the line segment A(1,-4), B(-2,2)?